Chapter 10

Two-point functions of local operators at one loop: design

This chapter defines the connected two-point function of two local operators in the UV-boundary formalism, proves that it exists at one loop at separated points, fixes its dependence on the separation by the dilatation component, lists how it changes under every scheme choice, derives the $\mathfrak{psl}(4|4)$ Ward identities it satisfies, gives the graph census for the scalar bilinears, the weight-$\frac{5}{2}$ fermion–scalar bilinears and the weight-3 sector of the Konishi anomaly, and evaluates the scalar bilinears exactly. The conjugation $X \mapsto X^\dagger$ is not used. Every statement carries its two-component form in the notation of [dictionary]; the only expressions written in words are the two-point function two-point$\big(O_1, O_2\big)$ and its tree part two-point$_{\text{tree}}\big(O_1, O_2\big)$ (Definition 10.1), and the unit

$$\tfrac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}$$

written out in full. $Q_1, S_1, D_1, K_1, P_1, M_1, R_1$ are the one-loop components of [odd, def:o-comp]; Q is the linear BV differential; spinor parameters are $\xi \in \Sigma$, $\eta \in \Sigma^*$ [odd, def:o-names]. References: [ch0n, label] are the manuscript chapters, [odd] is `step2/theory/odd_components_design.tex`, [dictionary] is `step2/notation/esw_to_weinberg_srednicki.tex`, [script] is `step2/theory/two_point_scalar_example.py` with output `two_point_example_results.json`, [census] is `two_point_census.json`; [conj], [ip], [delta_theory], [k_transfer] are `conjugation.tex`, `inner_product_design.tex` (checks `inner_product_checks.py`), `delta_theory.tex`, `k_transfer.tex` in the same folder. Revision 2026-10-29: the two-point function is the coefficient of $t_2 t_1$ (Definition 10.1), which fixes the sign of the fermionic tree values; the gauge-fixing law is a hypothesis; the special conformal identity uses the finite-endpoint local representative; the vanishing of the order-g entries at weight 3 is by colour and field content.

10.1 Definition and existence at one loop

Definition 10.1 (The two-point function)**.** Let $O_a = \int \omega_a\, f_a$, $a = 1, 2$, be homogeneous gauge-invariant local operators [ch05, def:c5-family] of degree 0 with $(Q + \{I, -\})O_a = 0$, Weyl weights w_1, w_2, and $\omega_1, \omega_2 \in \Omega^4_c(\mathbb{R}^4)$ with disjoint supports. Let t_1, t_2 be square-zero parameters of degree 0 and of the total parity of O_1, O_2, commuting with each other up to the Koszul sign, and put

$$J(\varepsilon) = I - F_0 - I^{CT}[\varepsilon] + t_1 O_1(\varepsilon) + t_2 O_2(\varepsilon), \qquad O_a(\varepsilon) = O_a - \partial_\delta I^{CT}[\varepsilon](O_a) + \hbar\,\varphi(O_a), \tag{10.1}$$

where F_0 is the arity-zero finite counterterm (at one loop $\frac{27}{16}\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)} \int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$ [ch04, thm:c4-finite] plus the s-closed choices of [odd, prop:o-arity0]), $\partial_\delta I^{CT}[\varepsilon](O_a)$ is the counterterm of [ch05, prop:c5-thmB] with all its Taylor components, those with the field content of O_a and those with

other field contents (length-changing), and $\hbar\varphi(O_a)$ is an ε-independent local functional (the finite part of $O_a(\varepsilon)$; $\varphi = 0$ is the minimal scheme). The two-point function of O_1 and O_2 is

$$\text{two-point}\big(O_1, O_2\big) := \lim_{\varepsilon\to 0}\ \partial_{t_1}\partial_{t_2}\, W\big(P(\varepsilon,\infty), J(\varepsilon)\big)\Big|_{e=0}, \tag{10.2}$$

the coefficient of t_2t_1 at the zero field, i.e. $(-1)^{p(O_1)p(O_2)}$ times the coefficient of t_1t_2 (p the total parity), a bilinear functional of (ω_1,ω_2); at tree level it is the free expectation $\hbar\,\partial_{P(0,\infty)}(O_1O_2)|_{e=0}$ with O_1 on the left, since $(t_1O_1)(t_2O_2) = t_2t_1O_1O_2$, the form in which the Ward identity (10.7) holds (for odd O_1, O_2 the coefficient of t_1t_2 is minus the expectation); its kernel at (x,y), $x\neq y$, is written $\text{two-point}\big(O_1,O_2\big)(x,y)$, and $\text{two-point}\big(O_1,O_2\big)(x)$ for $y=0$. Its tree part $\text{two-point}_{\text{tree}}\big(O_1,O_2\big)$ is the part of lowest order in the unit; for operators of equal length L it is the sum of the graphs with the two vertices O_1, O_2 and L edges, of order $\hbar^{L-1}$ [ch03, prop:c3-graphs]. The one-loop part is the part of relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$. With the global ghosts, $I - F_0 - I^{CT}[\varepsilon]$ is replaced by $I_{\mathfrak{psl}(4|4)} - F_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]$ and $O_a(\varepsilon)$ by the equivariant family [ch05, def:c5-Oeps]; the result is a cochain, the two-point cochain, whose arity-zero component is (10.2).

Two-component form: $\iota_w A = \frac14 w^{ij}A_{ij}$, $(w_1,w_2) = \frac14 w_1^{ij}w_{2ij}$, $A^{ij} = \frac12\varepsilon^{ijkl}A_{kl}$; the scalar bilinear is $\frac{1}{16}\int\omega\,(w_1^{ij}A_{ij}, w_2^{kl}A_{kl})$, the trace $\frac14\int\omega\,(A^{ij},A_{ij})$, and the propagator $\langle A_{ij}A_{kl}\rangle = 4(-,-)^{-1}\varepsilon_{ijkl}\,e^{-\varepsilon|p|^2}/|p|^2$ [dictionary, propagator, bilinears].

Lemma 10.2 (Elementary properties)**.** *(i) Constant terms of $O_a(\varepsilon)$ (for instance $-\frac{12\hbar\dim\mathfrak g}{16\pi^2\varepsilon}\int\omega$ in the trace bilinear [ch05, eq:c5-trace]) do not contribute to* (10.2)*.*

(ii) For every functional F, $(QF)|_{e=0} = 0$; hence Q-exact terms of the flow do not contribute to (10.2)*.*

(iii) $\text{two-point}\big(O_2,O_1\big)(y,x) = (-1)^{p(O_1)p(O_2)}\,\text{two-point}\big(O_1,O_2\big)(x,y)$*, p the total parity.*

(iv) $\{O_1(\varepsilon),O_2(\varepsilon)\}_\varepsilon$ and every K_ε edge joining a vertex supported on $\operatorname{supp}\omega_1$ *to one supported on* $\operatorname{supp}\omega_2$ *are bounded by a multiple of $\varepsilon^{-2}e^{-d^2/4\varepsilon}$, $d = \operatorname{dist}(\operatorname{supp}\omega_1,\operatorname{supp}\omega_2)$, and tend to* 0*.*

Proof. (i) $W(P, J + c\,t_1) = W(P,J) + c\,t_1$ for a constant c, by the definition of W [ch03, def:c3-W]; the t_1t_2 coefficient is unchanged. (ii) Q is the derivation induced by a linear operator on fields [C07, §6.2], so it preserves the polynomial degree and annihilates constants; the constant term of QF is 0. (iii) W is symmetric in its vertices; exchanging t_1, t_2 gives the Koszul sign. (iv) The scalar factor of K_ε is $(4\pi\varepsilon)^{-2}e^{-|x-y|^2/4\varepsilon}$ [ch03, def:c3-K]. □

Lemma 10.3 (Large distance)**.** *Let γ be a connected graph without legs, with vertices O_1 at* $\operatorname{supp}\omega_1$*, O_2 at* $\operatorname{supp}\omega_2$ *and internal vertices of I, F_0, $I^{CT}[\varepsilon]$, edges $P(\varepsilon,\infty)$. For a nonempty set S of internal vertices sent to infinity at a common scale R, the integrand decays as $R^{-4|S|-\sigma}$ with $\sigma =$ $\sum$(Weyl weights of the half-edges at the ends, outside S, of the edges leaving S). If $\sigma > 0$ for every S, the vertex integrations converge at large distance, uniformly in $\varepsilon\in(0,1]$. In particular they converge when no edge leaving S ends on a ghost c outside S; this holds for every graph of Section 10.5.*

Proof. An edge with half-edges of weights w, w' carries a kernel bounded by a multiple of $|z - z'|^{-(w+w')}$ for $|z-z'|\geq 1$, uniformly in $\varepsilon\leq 1$: $D_\varepsilon(r) = \frac{1-e^{-r^2/4\varepsilon}}{4\pi^2r^2}$ for $A\otimes A$ $(w+w'=2)$, dD_ε for $c\otimes A^*|_V$ $(0+3)$ and $\not dD_\varepsilon$ for $\lambda\otimes\lambda$ $(\frac32+\frac32)$ [ch03, prop:c3-P]. A vertex of weight 4 (fields plus derivatives [ch04, lem:c4-weights]) with all its derivatives acting on edges contributes, together with its edges, $R^{-(\text{sum of weights of its half-edges and derivatives})} = R^{-4}$; edges internal to S are counted at both ends, edges leaving S contribute in addition the weight of their far end. Hence the integrand

is $O(R^{-4|S|-\sigma})$ against the measure $R^{4|S|-1}dR$. Finite counterterm vertices of weight < 4 (the ε^{-1} mass term) decay faster. The weights of $A, \lambda, A^*, \lambda^*, c^*$ are positive; a zero weight occurs only for c. □

Lemma 10.4 (Fourier transforms at separated points)**.** *Let a translation-invariant two-point kernel have the momentum-space form $h(\varepsilon|q|^2)\,|q|^{-2k}$, $k \in \{0,1\}$, with h smooth on $(0,\infty)$, bounded by a polynomial in $1+|\log y|$, and $h(y) = \sum_{j=0}^{2} c_j \log^j y + O\big(y(1+|\log y|)^2\big)$ as $y \to 0$. Then at $x \neq 0$, as $\varepsilon \to 0$, its position-space kernel converges to the Fourier transform of $\sum_j c_j(\log\varepsilon + \log|q|^2)^j|q|^{-2k}$, given by*

$$\int \tfrac{d^4q}{(2\pi)^4} e^{i(q,x)} \log|q|^2 = -\frac{1}{\pi^2|x|^4}, \qquad \int \tfrac{d^4q}{(2\pi)^4} e^{i(q,x)} \log^2|q|^2 = \frac{2}{\pi^2|x|^4}\big(\log|x|^2 + 2\gamma_E - 1 - 2\log 2\big),$$
$$\int \tfrac{d^4q}{(2\pi)^4} e^{i(q,x)} \frac{1}{|q|^2} = \frac{1}{4\pi^2|x|^2}, \qquad \int \tfrac{d^4q}{(2\pi)^4} e^{i(q,x)} \frac{\log|q|^2}{|q|^2} = -\frac{\log|x|^2 + 2\gamma_E - 2\log 2}{4\pi^2|x|^2},$$

and constants (polynomials in q) contribute 0 at $x \neq 0$.

Proof. The four transforms are the coefficients of a^1, a^2 in $\int \frac{d^4q}{(2\pi)^4} e^{i(q,x)}|q|^{2a} = \frac{4^a\Gamma(2+a)}{\pi^2\Gamma(-a)}|x|^{-4-2a}$ and $\int \frac{d^4q}{(2\pi)^4} e^{i(q,x)}|q|^{2a-2} = \frac{4^a\Gamma(1+a)}{4\pi^2\Gamma(1-a)}|x|^{-2-2a}$ at $x \neq 0$, continued in a from the region of absolute convergence (an analytic continuation in the exponent of $|q|$, not a change of dimension); [script, C] expands them exactly. The remainder $r(y) = h(y) - \sum_j c_j \log^j y$ gives the kernel $\varepsilon^{-2+k}\hat{r}_k(x/\sqrt{\varepsilon})$ with $\hat{r}_k$ the transform of $r(|q|^2)|q|^{-2k}$; since $r(|q|^2)|q|^{-2k}$ is smooth for $q \neq 0$, of polynomial growth, and $O(|q|^{2-2k}\log^2|q|)$ at $q = 0$, $\hat{r}_k(X) = O(|X|^{-6+2k}\log^2|X|)$ at large $|X|$, so the kernel is $O(\varepsilon\log^2\varepsilon)$ at fixed $x \neq 0$. □

Proposition 10.5 (Existence at one loop)**.** *For O_1, O_2 as in Definition 10.1, the limit* (10.2) *exists at $x \neq y$ through relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}$, including the length-changing entries. It is a sum over the classes of graphs:*

- *(T) the tree part: O_1 and O_2 joined by edges (equal lengths), or joined through one or more vertices of I (different lengths, Proposition 10.6);*
- *(SE) a line of the tree carrying a one-loop one-particle-irreducible two-point subgraph of I together with the vertex $-I^{CT}[\varepsilon]$ (and $-F_0$);*
- *(V) graphs containing a one-loop subgraph through O_a (the graphs of [ch05, prop:c5-census] with their legs contracted into the rest), together with the trees carrying the vertex $-\partial_\delta I^{CT}[\varepsilon](O_a)$, $a = 1, 2$; when a subgraph through O_1 and one through O_2 share edges (exchange graphs) both subtractions occur;*
- *(F) trees with one vertex $\hbar\varphi(O_a)$.*

Proof. By [ch03, prop:c3-graphs] the order counted from the tree part is $b_1(\gamma) + \sum_v \deg v$ minus its value at the tree part, so at relative order one either one more cycle appears with vertices of I, or one vertex of order $\hbar$ ($-I^{CT}[\varepsilon]$, $-F_0$, $-\partial_\delta I^{CT}[\varepsilon](O_a)$, $\hbar\varphi(O_a)$) replaces a tree position. UV: in position space the integrand is singular only where a set of vertices collapses to a point; a collapsing set containing both O_1 and O_2 is excluded by $\mathrm{dist}(\mathrm{supp}\,\omega_1, \mathrm{supp}\,\omega_2) > 0$. The remaining collapsing sets at this order are one-loop subgraphs: those without O_a are the one-particle-irreducible subgraphs of I, whose singular parts are cancelled by the trees with $-I^{CT}[\varepsilon]$ [ch04, lem:c4-factor, prop:c4-limit]; those containing O_a are cancelled by $-\partial_\delta I^{CT}[\varepsilon](O_a)$ [ch05, prop:c5-thmB]. The complementary

graph, obtained by contracting the subgraph to a point, is a tree with nonzero edge momenta at $x \neq y$, so the factorisation of [ch04, lem:c4-factor] applies; two collapsing sets at O_1 and at O_2 are disjoint in position space, and their subtractions factorise. Edges of the trees tend to $P(0,\infty)$ by Lemma 10.4 and [ch05, prop:c5-thmB, first item]. Large distance: Lemma 10.3. The order-$\hbar$ vertices $-F_0$, $\hbar\varphi(O_a)$ enter as tree vertices with limits. Every remaining ε-dependence of the subtracted graphs is of the form of Lemma 10.4, by dimensional analysis: each graph is $|x-y|^{-w_1-w_2}$ times a function of $|x-y|^2/\varepsilon$ and of $\log\varepsilon$ through the counterterms only. □

Proposition 10.6 (Length-changing entries)**.** *Let O_L, O_{L+1} be gauge-invariant operators of lengths L, $L+1$. The entry of order g (one cubic vertex of I) is the sum of the graphs in which the cubic vertex carries two legs of O_{L+1} and one leg of O_L; its only subdivergence is the one-loop subgraph formed by O_{L+1} and the cubic vertex, removed by the length-changing component of $\partial_\delta I^{CT}[\varepsilon](O_{L+1})$. It vanishes unless (a) the two legs of O_{L+1} at the cubic vertex carry an antisymmetric colour tensor (the bracket), and (b) the third leg of the cubic vertex matches a leg of O_L. For the operators of Section 10.5 (Lorentz scalars of weight ≤ 3, and O_ζ) every order-g entry vanishes: the cubic vertices with two scalar legs have a third leg $A|_V$, which no gauge-invariant Lorentz-scalar bilinear of weight ≤ 3 contains, and the Yukawa vertex carries one scalar leg, so a scalar trilinear meets it with one leg and its two remaining legs are joined to each other, which gives the colour factor $\big(Z,[-,-]\big)$ on $(-,-)^{-1}$, i.e. 0 [ch05, lem:c5-colour]. For lengths L, $L+1$ the next graphs have an odd number of cubic vertices (three, or one cubic and one quartic), since the number of half-edges must be even. When the longer operator carries a bracket, the order-g entry is itself of relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ with respect to the tree part of the shorter pair (the coupling sits in $(-,-)$); the example used below is O_ζ with the trilinear part of $\{S^{(1)}(\xi),F_n\}$, colour factor $-\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)/(X,Y)\cdot\dim\mathfrak g$ [census].*

Proof. A connected graph with O_L, O_{L+1} and one cubic vertex has $(L+L+1+3)/2 = L+2$ edges. If the cubic vertex carries two legs of O_L and one of O_{L+1}, the two legs of O_L meet $(-,-)$ (the colour factor of a gauge-invariant bilinear contracted on two legs is symmetric) with the antisymmetric cubic colour factor, giving 0; if it carries one leg of each and one further leg, that leg must join O_L or O_{L+1}, the first case being the previous one and the second one in which O_{L+1} has a self-contraction, 0 by [ch05, lem:c5-colour]. The subgraph $\{O_{L+1},\text{cubic vertex}\}$ is a cycle of two edges, the only collapsing set. The vanishing statements are case checks of (a), (b) with the vertex list of [ch01, def:c1-vertex]. □

10.2 Dependence on the separation

For a homogeneous operator O write $Z(O)$ for the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon](O)$, and $\{N,O\}$ for its bracket with the field-counting functional $N = 2\int(A,A^*) - 4\int(\lambda^*,\lambda) - \int(c^*,c)$ [ch06, lem:c6-count]. The dilatation component of [ch06, thm:c6-dil, cor:c6-dilcomp], identified with the equivariant cochain along $\Delta = 2E$ (Part D, eq:c6-step2), is, for gauge-invariant cocycles,

$$\Delta\cdot O := 2\big((E+w_O)O\big) + D_1(O), \qquad D_1(O) = 4\Big(Z(O) - \tfrac{\hbar\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\{N,O\}\Big), \tag{10.3}$$

modulo s-exact local functionals; for the trace bilinear $D_1(O) = -48\frac{\hbar\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,O$, for the traceless ones 0 [ch06, thm:c6-dil]. In the generator $-E$ of [SW, (3)] every statement below holds with Δ replaced by $-\frac12\Delta$.

Theorem 10.7 (Scaling identity)**.** *Let O_1, O_2 be as in Definition 10.1, with F_0 of weight 4 and φ preserving the Weyl weight. Through relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$,*

$$\begin{aligned}&\big(x\cdot\partial_x+y\cdot\partial_y+w_1+w_2\big)\text{two-point}\big(O_1,O_2\big)(x,y)\\&\qquad=-\tfrac12\Big[\text{two-point}_{\text{tree}}\big(D_1(O_1),O_2\big)+\text{two-point}_{\text{tree}}\big(O_1,D_1(O_2)\big)\Big](x,y).\end{aligned}\tag{10.4}$$

Equivalently, $\text{two-point}\big(\Delta\cdot O_1,O_2\big)+\text{two-point}\big(O_1,\Delta\cdot O_2\big)=0$*, with Δ acting on the labels O_a, its classical part through ω_a (the dilatation Ward identity).*

Proof. On numbers σ_s acts trivially: $(\sigma_sF)(0)=F(\sigma_s^{-1}0)=F(0)$ [ch06, def:c6-sigma]. By [ch06, prop:c6-scaling], $\partial_{t_2}\partial_{t_1}W(P(\varepsilon,\infty),J(\varepsilon))|_{e=0}=\partial_{t_2}\partial_{t_1}W(P(s^2\varepsilon,\infty),\sigma_sJ(\varepsilon))|_{e=0}$. By [ch06, lem:c6-ct] and $\sigma_sI=I$, $\sigma_sF_0=F_0$,

$$\sigma_sJ(\varepsilon)=I-F_0-I^{CT}[s^2\varepsilon]+2\log s\,Y+t_1\big((\sigma_sO_1)(s^2\varepsilon)+2\log s\,\sigma_sZ(O_1)\big)+t_2\big(\cdots\big),$$

Y the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$. Letting $\varepsilon\to0$ (Proposition 10.5) and differentiating at $s=1$: by the linearity in ω_a and [ch06, lem:c6-data], $s\frac{d}{ds}|_{s=1}$ of the two-point function of σ_sO_1,σ_sO_2 is $\iint h_1(x)h_2(y)(x\cdot\partial_x+y\cdot\partial_y+w_1+w_2)\text{two-point}\big(O_1,O_2\big)(x,y)$ for $\omega_a=h_a\,\mathrm{dvol}$, since the two-point function of σ_sO_a is $s^{w_1+w_2}\iint h_1h_2\,\text{two-point}\big(O_1,O_2\big)(sx,sy)$. The $2\log s$ terms are of order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ and enter through tree graphs:

$$\begin{aligned}0=&(x\cdot\partial_x+y\cdot\partial_y+w_1+w_2)\text{two-point}\big(O_1,O_2\big)\\&+2\text{two-point}_{\text{tree}}\big(Z(O_1),O_2\big)+2\text{two-point}_{\text{tree}}\big(O_1,Z(O_2)\big)+2\big(\text{tree insertion of }Y\big).\end{aligned}$$

By [ch06, eq:c6-fieldren] $Y=\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}(Q+\{I,-\})N$, and the two-point form of [ch06, lem:c6-bvexact] (same proof with two operators; Lemma 10.2 (ii) removes the Q-term at $e=0$, and the terms built from $(Q+\{I,-\})O_a$ vanish) gives the tree insertion $-\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\big(\text{two-point}_{\text{tree}}\big(\{N,O_1\},O_2\big)+\text{two-point}_{\text{tree}}\big(O_1,\{N,O_2\}\big)\big)$. With (10.3) this is (10.4). The equivalent form: the classical part $2(E+w_a)$ of Δ on O_a acts on the kernel as $2(x\cdot\partial_x+w_1)$, resp. $2(y\cdot\partial_y+w_2)$. □

Corollary 10.8 (Form of the two-point functions of scaling operators)**.** *Let O_a, $a=1,\dots,n$, be Lorentz-scalar cocycles of common weight w with $D_1(O_a)=\sum_b(D_1)_{ab}O_b$ modulo s-exact terms,* $\text{two-point}_{\text{tree}}\big(O_a,O_b\big)($ *$C_{ab}|x|^{-2w}$. Then through relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$*

$$\text{two-point}\big(O_a,O_b\big)(x)=|x|^{-2w}\Big[C_{ab}+\big(\textit{one-loop finite matrix}\big)_{ab}-\tfrac14\big(D_1C+CD_1^T\big)_{ab}\log\frac{|x|^2}{\ell^2}\Big],\tag{10.5}$$

where ℓ is the unit of length in which ε is measured: the counterterms $\log\varepsilon$ of [ch05, thm:c5-Oeps] and [ch04, thm:c4-log] are $\log(\varepsilon/\ell^2)$. The one-loop finite matrix is the one-loop part at $|x|=\ell$. Changing $\ell\mapsto\lambda\ell$ is the scheme change $\hbar\varphi(O)=Z(O)\log\lambda^2$ (type (F3)) together with $F_0\mapsto F_0-Y\log\lambda^2$ (type (F0), Y the coefficient of $\log\varepsilon$ in $I^{CT}[\varepsilon]$); it adds $\frac14(D_1C+CD_1^T)|x|^{-2w}\log\lambda^2$, so that (10.5) holds with $\lambda\ell$ in place of ℓ and the same finite matrix.

Proof. Translations and rotations (Proposition 10.9) give $\text{two-point}\big(O_a,O_b\big)(x,y)=g_{ab}(|x-y|)$. With $g_{ab}=r^{-2w}(c_{ab}+e_{ab}\log r^2)$, $(r\partial_r+2w)g_{ab}=2e_{ab}r^{-2w}$, and (10.4) gives $2e=-\frac12(D_1C+CD_1^T)$. Every ratio $|x|^2/\varepsilon$ produced by the graphs is dimensionless; $\log\varepsilon$ alone occurs only through the counterterms, so the only scale is the unit ℓ fixed with them. □

Proposition 10.9 (Translations, rotations, R-symmetry)**.** *H, Q^{GF}, I, $I^{CT}[\varepsilon]$ and F_0 are invariant under translations,* $\mathrm{Spin}(4,\mathbb{C})$ *and* $\mathrm{Spin}(6,\mathbb{C})$ *[ch04, thm:c4-quadratic, proof; ch07, prop:c7-cov]. With φ covariant,* $\text{two-point}\big(g\cdot O_1, g\cdot O_2\big) = \text{two-point}\big(O_1,O_2\big)$ *for g in these groups acting on the labels; in particular* $\text{two-point}\big(O_1,O_2\big)(x,y)$ *depends on $x-y$, and on its length for Lorentz scalars. In the scheme $P_1 = M_1 = R_1 = 0$ [odd, thm:o-invariants], these are the Ward identities of Theorem 10.12 along translations, rotations and $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$.*

Proof. The regulator and all vertices are covariant, so the whole graph sum is; the limit preserves covariance. □

10.3 Scheme covariance

Proposition 10.10 (Transformation laws)**.** *Through relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$, with* $\mathrm{ins}(G;O_1,O_2)$ *the tree graphs with one vertex G (sum over the lines of the tree):*

(F0) $F_0 \mapsto F_0 + \hbar G$*, G local of weight 4 and degree 0:* $\text{two-point}\big(O_1,O_2\big) \mapsto \text{two-point}\big(O_1,O_2\big) - \mathrm{ins}(\hbar G;O_1,O_2)$*. For* $\hbar G = (Q+\{I,-\})\phi$ *(ϕ odd, a field redefinition):* $\mapsto \text{two-point}\big(O_1,O_2\big) + \text{two-point}_{\text{tree}}\big(\{\phi,O_1\},O_2\big) + \text{two-point}_{\text{tree}}\big(O_1,\{\phi,O_2\}\big)$*. For the forced term* $\frac{27}{16}\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V$ *the change is 0 whenever no tree line is an $A|_V$ line, and also when the $A|_V$ lines end on vertices built from F_A (transversality, ι_p of the vertex function vanishes). For the coupling direction $\hbar G = a(S - S^{(2)})$ at arity zero, a S: every edge is multiplied by $1-a$ and every vertex of I by $1+a$, so a tree with L edges and m vertices of I is multiplied by $1+(m-L)a$.*

(F3) $O_a(\varepsilon) \mapsto O_a(\varepsilon) + \hbar\varphi(O_a)$*:* $\text{two-point}\big(O_1,O_2\big) \mapsto \text{two-point}\big(O_1,O_2\big) + \text{two-point}_{\text{tree}}\big(\hbar\varphi(O_1),O_2\big) + \text{two-point}_{\text{tree}}\big(O_1,\hbar\varphi(O_2)\big)$*.*

(F1/F2) arity-one and arity-two finite counterterms $F^{(1)}, F^{(2)}$: no change of $\text{two-point}\big(O_1,O_2\big)$*, which is built from the arity-zero interaction; they change L_1 by $[s_{\mathfrak{psl}(4|4)}, \mathrm{ad}_\phi]$ or by an s-exact choice [odd, prop:o-laws (a)], and the Ward identities of Theorem 10.12 are invariant (Corollary 10.14).*

(gauge) (hypothesis) if two gauge-fixing operators are joined by a family Q_t^{GF} with the properties of [odd, prop:o-laws (c)], the derivative in t is a sum of changes of types (F0) with $\hbar G$ exact, (F3), and the coupling direction; this is the proof plan of [odd, prop:o-laws (c)]. For the Feynman operator of the minimal complex, with $(Q^{GF})^2 \neq 0$, a family of this type ending at a nilpotent gauge fixing with a Laplacian H needs an enlarged field space [delta_theory, rem:d-gf], and the statement is not established.

Proof. (F0) At order $\hbar$ the added vertex $-\hbar G$ enters linearly through trees. For $\hbar G = (Q+\{I,-\})\phi$ apply the two-point form of [ch06, lem:c6-bvexact] used in the proof of Theorem 10.7, with the opposite sign of the vertex. The forced term is $\int \mathrm{dvol}\,(d^*A|_V, d^*A|_V)$, whose vertex function on two $A|_V$ legs of momenta $\pm p$ is $2(\iota_p A(p), \iota_p A(-p))$ [ch04, eq:c4-pw]: inserted in a line it contracts ι_p with the vertex functions at the ends. For the coupling direction, $S \mapsto (1+a)S$ multiplies the quadratic form by $1+a$, its inverse P by $1-a$ at this order. (F3) linearity. (F1/F2) the arity-zero component of the two-point cochain contains no ghost vertex. (gauge) under its hypothesis, [odd, prop:o-laws (c), proof plan]: the Duhamel term at $t=\varepsilon$ is local and of the listed types; the term $(Q\otimes 1 + 1\otimes Q)$ of a kernel gives Q-exact terms, zero at $e=0$ (Lemma 10.2 (ii)), up to brackets with O_a, which are of type (F3). □

Corollary 10.11 (Invariant content)**.** *(i) The coefficient of* $\log|x|^2$ *in* (10.5)*, $-\frac{1}{4}(D_1C + CD_1^T)$, is invariant under (F0), (F1/F2), (gauge) under its hypothesis, and under (F3) with φ weight-preserving [odd, thm:o-invariants (ii)].*

(ii) *The finite matrix is not invariant: (F3) shifts it by $\hbar(\varphi C + C\varphi^T)$ (with $\varphi(O_a) = \sum_b \varphi_{ab} O_b$), (F0) by the insertion, the unit ℓ by Corollary 10.8.*

(iii) *If (F3) is restricted to $\mathfrak{psl}(4|4)$-equivariant finite parts (the map ψ of [odd, prop:o-laws (b)] commuting with the classical $\mathfrak{psl}(4|4)$ action on the multiplet), φ is constant on each irreducible multiplet; then differences of the finite parts of the normalised two-point functions between members of one multiplet are invariant, and they are fixed by the Ward identities (Theorem 10.12).*

(iv) *Orthogonality:* $\text{two-point}\big(O_a, O_b\big) = 0$ *through relative order* $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}$ *when O_a, O_b have different eigenvalues of D_1 at equal weight, up to the tree mixing C_{ab} (zero if C is block-diagonal in the eigenspaces).*

Proof. (i) By (10.4) the coefficient is built from D_1 and the tree part; D_1 changes by s-exact terms under (a)–(d) [odd, thm:o-invariants (ii)], which have vanishing two-point functions with cocycles (Corollary 10.13). (ii) Proposition 10.10. (iii) Schur's lemma for the irreducible multiplet under the classical action. (iv) Diagonalise D_1; (10.4) for O_a, O_b with eigenvalues $d_a \neq d_b$ and $C_{ab} = 0$ gives $(r\partial_r + 2w) g_{ab} = 0$ with g_{ab} of relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}$; Theorem 10.12 along K (Example 10.17) forces $g_{ab} = 0$. □

10.4 Ward identities

For a generator $x \in \mathfrak{psl}(4|4)$ and a local operator O with local representative $L(O)$ [odd, def:o-comp] put

$$x \cdot O := L(O)^{(1)}(x) = \{S^{(1)}(x), O\} + \hbar L_1(O)^{(1)}(x). \tag{10.6}$$

The classical part is position dependent: for a conformal Killing field v and a Lorentz-scalar density f of weight w, $\{S^{(1)}(v), \int \omega\, f\} = \int \omega \left(v(f) + \frac{w}{4}(\mathrm{div}\, v) f\right)$ [ch07, lem:c7-weights], acting on the kernel as $v(x) \cdot \partial_x + \frac{w}{4} \mathrm{div}\, v(x)$; for $\Delta = 2E$ this is $2(x \cdot \partial_x + w)$, for K_{v_1}, $v(x) = (x,x)v_1 - 2(v_1, x)x$, $\mathrm{div}\, v = -8(v_1, x)$, two-component $v^m = x^2 b^m - 2(b \cdot x)x^m$, $\mathrm{div}\, v = -8\, b \cdot x$, $b = v_1$; for $\eta \in \Sigma^*$ the Killing spinor $\xi + \rho(x)\eta$, two-component $\epsilon_\alpha{}^i(x) = \xi_\alpha{}^i - i x_m \sigma^m_{\alpha\dot\alpha} \bar\eta^{\dot\alpha i}$, $\bar\epsilon_{\dot\alpha i}(x) = \bar\xi_{\dot\alpha i} + i x_m \eta^\alpha{}_i \sigma^m_{\alpha\dot\alpha}$ [odd, def:o-names], together with the term in 4η [ch02, eq:c2-S1S].

Theorem 10.12 (Two-point Ward identity)**.** *Assume*

(H1) *(no anomaly) the equivariant one-loop master equation holds in arities* 0 *and* 1 *along x after finite counterterms $F_{\mathfrak{psl}(4|4)}$;*

(H2) *(local representatives) O_a have local representatives with $L(O_a)^{(0)} = 0$ through order $\hbar$, and the flow of the t_a-linear part of the defect vertex converges to $L(O_a)[\infty]$ as a distribution in the leg positions, tested against sections smooth near* $\mathrm{supp}\, \omega_a$*;*

(H3) *(large distance) the arity-one graphs along x converge at large distance: $x \in \Sigma \oplus \Sigma^*$, Δ, translations, $\mathfrak{so}(V)$, $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ [odd, thm:o-ir (i)];*

(H4) $\mathrm{supp}\, \omega_1 \cap \mathrm{supp}\, \omega_2 = \emptyset$.

Then through relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}$

$$\text{two-point}\big(x \cdot O_1, O_2\big) + (-1)^{p(x)p(O_1)}\, \text{two-point}\big(O_1, x \cdot O_2\big) = 0. \tag{10.7}$$

Proof. Let $\mathscr{G}$ be the two-point cochain (Definition 10.1) and J_t the equivariant family. By [ch03, thm:c3-uv] and [ch06, eq:c6-uvobs], $(d_{CE}+Q)W(P(\varepsilon,\infty),J_t)$ is the flow of the defect vertex $(d_{CE}+Q)J_t+\frac{1}{2}\{J_t,J_t\}_\varepsilon+\hbar\Delta_\varepsilon J_t$. Its t-independent part tends to 0 in arities $0,1$ by (H1); its t_1t_2 part is $\{O_1(\varepsilon),O_2(\varepsilon)\}_\varepsilon\to 0$ (Lemma 10.2 (iv), (H4)); its t_a-linear parts are the operator defects of [odd, eq:o-defect], whose flows in the presence of the vertex $t_bO_b(\varepsilon)$ converge by (H2), (H3) to the two-point cochains of $L(O_1)$ with O_2 and of O_1 with $L(O_2)$: the singular regions are those of the defect at O_a alone (Proposition 10.5, proof), and the lines to O_b are smooth near $\operatorname{supp}\omega_a$. Take the t_2t_1 coefficient at $e=0$. On the left, the Q-part vanishes (Lemma 10.2 (ii)); d_{CE} has trivial coefficients [ch02, def:c2-dce], so it maps arity 0 to 0 and arity k to arity $k+1$; the arity-one component of $\mathscr{G}$ at $e=0$ is a number of degree -1, hence 0. So the left side vanishes in arities 0 and 1. On the right, in arity one along x, the terms are two-point$\big(L(O_1)^{(1)}(x),O_2\big)$, two-point$\big(O_1,L(O_2)^{(1)}(x)\big)$ with the Koszul sign of moving the ghost of x past t_1O_1, and terms containing $L(O_a)^{(0)}$, which vanish by (H2). □

Corollary 10.13 (Exact operators)**.** *Under (H1), (H2), (H4), for every local A of degree -1 with a local representative,* two-point$\big(L(A)^{(0)},O_2\big)=0$. *The two-point function descends to the cohomology of $L^{(0)}=(Q+\{I,-\})+\hbar L_1^{(0)}$ on local operators; in particular it is independent of the s-exact ambiguity of $x\cdot O$, and of D_1, K_1 modulo s-exact terms.*

Proof. The arity-zero part of the proof of Theorem 10.12 applied to the pair (A,O_2): $0=$ two-point$\big(L(A)^{(0)},O_2\big)\pm$ two-point$\big(A,L(O_2)^{(0)}\big)$, and $L(O_2)^{(0)}=0$. □

Corollary 10.14 (Covariance of the identity)**.** *Under the scheme changes (a)–(d) of [odd, prop:o-laws] and (F0), (F3) of Proposition 10.10,* two-point$\big(-,-\big)$ *and $x\cdot(-)$ change so that* (10.7) *is preserved; finite parts φ of operators are compatible with* (10.7) *exactly when $[\,x\cdot,\varphi\,]$ is s-exact on the operators considered.*

Proof. Theorem 10.12 holds in every scheme satisfying (H1)–(H4). For (F3) directly: $x\cdot$ changes by $[s_{\mathfrak{psl}(4|4)},\psi]$ [odd, prop:o-laws (b)], whose arity-one part on a cocycle is $[\,x\cdot,\psi]$ plus s-exact terms; by Proposition 10.10 the left side of (10.7) changes by two-point$_{\text{tree}}\big(\hbar\varphi(x\cdot O_1)-x\cdot\hbar\varphi(O_1),O_2\big)+(\cdots)$ plus the image of (10.7) under φ, and Corollary 10.13 removes the s-exact terms. □

Proposition 10.15 (The special conformal identity from the special supercharges)**.** *Let $\eta_1,\eta_2\in\Sigma^*$, $v_1=\Gamma(\eta_1,\eta_2)|_V$, and let K_1 be defined by [odd, def:o-K]. Assume (H1)–(H4) for $x=\eta_1,\eta_2$, the arity-two components of the cochain identity [odd, thm:o-cocycle] at (η_1,η_2) on O_1,O_2 and on $\eta_a\cdot O_b$, and the existence and T-independence of the local representative $L_1^{(2)}(\eta_1,\eta_2)$ at a finite endpoint T [k_transfer, thm:ktr-defect (v)]; the arity-two flows themselves grow as $\log T$ when $\Gamma(\eta_1,\eta_2)|_V\neq 0$, and only the local representative enters. Then*

$$\text{two-point}\big(K_{v_1}\cdot O_1,O_2\big)+\text{two-point}\big(O_1,K_{v_1}\cdot O_2\big)=0$$

with $K_{v_1}\cdot O=\{S^{(1)}(K_{v_1}),O\}+K_1(O)$; no arity-one graph along K_{v_1}, in particular no floating insertion of $S^{(1)}(K_{v_1})$ with its borderline decay $|y|^{-4}$ [odd, prop:o-ir], enters.

Proof. Write $\rho(x)O:=x\cdot O$. The arity-two component of [odd, thm:o-cocycle] at (η_1,η_2) on a cocycle O reads $\rho(\eta_1)\rho(\eta_2)O+\rho(\eta_2)\rho(\eta_1)O=\pm\rho([\eta_1,\eta_2])O+L^{(0)}\big(L^{(2)}(\eta_1,\eta_2)O\big)$, the sign fixed by [ch02, eq:c2-dce]; [odd, def:o-K] defines K_1 exactly through this relation, with $[\eta_1,\eta_2]=-K_{v_1}$ [ch02, prop:c2-special]. For O_1,O_2 of even parity, (10.7) for the odd η_a gives two-point$\big(\eta_1\eta_2O_1,O_2\big)=$ two-point$\big(\eta_2O_1,\eta_1O_2\big)=-$two-point$\big(O_1,\eta_2\eta_1O_2\big)$ (first with the

odd operator $\eta_2 O_1$, sign -1, then with O_1, sign $+1$), and likewise with $1 \leftrightarrow 2$. Adding, two-point$\big((\eta_1\eta_2+\eta_2\eta_1)O_1, O_2\big) = -$two-point$\big(O_1, (\eta_1\eta_2+\eta_2\eta_1)O_2\big)$; the $L^{(0)}$-exact terms drop by Corollary 10.13, and $\rho([\eta_1,\eta_2]) = -\rho(K_{v_1})$ appears linearly on both sides. The pairs $\eta_1\eta_2$ span $\mathrm{Sym}^2\Sigma^*$, whose image under $\Gamma|_V$ is V, so every v_1 occurs. □

Remark 10.16 (Finite transfer endpoint)**.** With $P(\varepsilon, T)$ [DESIGN_STEP2, R7] the two-point function at scale T satisfies (10.7) up to the brackets $\{-,-\}_T$ between $\operatorname{supp}\omega_1$ and $\operatorname{supp}\omega_2$, bounded by $T^{-2}e^{-d^2/4T}$ times polynomials, which vanish as $T \to \infty$; at finite T the arity-one graphs along K_{v_1} converge, and if K_1 is defined at finite T and shown T-independent (R7), Theorem 10.12 applies along K_{v_1} directly and agrees with Proposition 10.15 by [odd, prop:o-Kconsistent].

Example 10.17 (The special conformal identity on scalar operators)**.** Let O_1, O_2 be Lorentz-scalar operators of weight w with $D_1(O_a) \equiv 2\gamma_a O_a$ and, by [odd, thm:o-KD], $K_1(O_a) \equiv \frac{1}{4}\operatorname{div} v\,\gamma_a O_a$. Write two-point$\big(O_1, O_2\big)(x,y) = g(|x-y|^2)$. Since $(v(x)\cdot\partial_x + v(y)\cdot\partial_y)|x-y|^2 = -2\big((v_1,x)+(v_1,y)\big)|x-y|^2$,

$$\text{two-point}\big(K_{v_1}\cdot O_1, O_2\big)+\text{two-point}\big(O_1, K_{v_1}\cdot O_2\big) = -2(v_1,x)\big(rg'+(w+\gamma_1)g\big)-2(v_1,y)\big(rg'+(w+\gamma_2)g\big), \quad r=|x-y|^2,$$

so the identity holds for all v_1, x, y exactly when $g=0$ or $\gamma_1=\gamma_2=\gamma$ and $r\,g' = -(w+\gamma)g$; through relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$, $g = C\,r^{-w}\big(1+(\text{finite})-\gamma\log r\big)$, the form of Corollary 10.8 with $D_1 = 2\gamma$. Two-component: $v^m\partial_m + \frac{w+\gamma}{4}\partial_m v^m$ with $\partial_m v^m = -8\,b\cdot x$. The finite constant is not constrained by K: it is the scheme freedom (F3).

Proposition 10.18 (Selection rules and the nontrivial odd identities)**.** *Let c be the central element of* $\mathrm{Spin}(6;\mathbb{C}) \cong SL(4,\mathbb{C})$ *acting by i on W, $-i$ on W^*, -1 on $W_{\mathbb{R}}^{\perp}$; it preserves the action and the regulator, so* two-point$\big(O_1, O_2\big) = 0$ *unless the product of the eigenvalues of c on O_1, O_2 is* 1. *The propagator joins ${\lambda_\alpha}^i$ only to $\bar\lambda_{\dot\beta j}$. On fields $\xi\cdot$ ($\xi \in S_+\otimes W$) multiplies the eigenvalue by $-i$. Eigenvalues: the scalar bilinears* 1*; $O_\zeta = \int\omega\,(\lambda, \rho(A|_{W_{\mathbb{R}}^{\perp}})\zeta)$ with $\zeta \in S_+\otimes W$ (two-component $\int\omega\,({\lambda_\alpha}^j, A_{ji})\zeta^{\alpha i}$): $-i$; with $\bar\zeta \in S_-\otimes W^*$ (two-component $-\int\omega\,(\bar\lambda_{\dot\alpha j}, A^{ji})\bar\zeta^{\dot\alpha}{}_i$): i; B_w, $\bar F_w$ [odd, eq:o-BF]: -1; $F_n = \int\omega\,(\lambda^{\alpha i}n_i, {\lambda_\alpha}^j n_j)$, $n \in W^*$, and $\bar B_n = \int\omega\,([A_{ik}, A^{kj}n_j], A^{il}n_l)$: -1. Hence the nontrivial identities* (10.7) *along $\xi = u\otimes w$ among these operators are*

- *(a)* $\big(O,\ O_{\bar\zeta}\big)$, *$O$ a scalar bilinear: relating the tree and one-loop parts of* two-point$\big(O_\xi, O_{\bar\zeta}\big)$ *to ∂* two-point$\big(O, O\big)$*;*
- *(b)* $\big(O_\zeta,\ F_n\big)$ *and* $\big(O_\zeta,\ \bar B_n\big)$*: the Konishi-anomaly identities, relating c_K, c'_B of [odd, prop:o-konishi] to the weight-*3 *two-point functions.*

The η-identities are the same with the Killing spinor in place of ξ.

Proof. c lies in $\mathrm{Spin}(6;\mathbb{C})$, which commutes with H and Q^{GF} and preserves I, $I^{CT}[\varepsilon]$, F_0 [ch04, thm:c4-quadratic, proof]; it acts on $\lambda \in S_+\otimes W \oplus S_-\otimes W^*$ by $(i,-i)$ and on $\iota_w A$ by -1, since $W_{\mathbb{R}}^{\perp} \cong \wedge^2 W$. On fields, $\delta_\xi A \propto \Gamma(\xi,\lambda)$ maps eigenvalue -1 to i, and $\delta_\xi\lambda \propto \rho(F_A)\xi$ maps i (resp. $-i$) to 1 or -1 products accordingly; a direct count gives the factor $-i$. The pairs listed are those with product i (so that $\xi\cdot O_1$ times O_2 has product 1); the propagator rule removes $(O_\zeta, \xi\cdot O)$-type tree terms with two undotted λ. □

Proposition 10.19 (Content of the identities (a) and (b))**.** *(a) Let $K = \int\omega\,(A|_{W_{\mathbb{R}}^{\perp}}, A|_{W_{\mathbb{R}}^{\perp}})$ (two-component $\frac{1}{4}\int\omega\,(A^{ij}, A_{ij})$). Classically $\{S^{(1)}(\xi), K\} = \mp 2\,O_{\zeta=\xi}$ (sign of [ch01, def:c1-bracket], engine E0). With* two-point$\big(K,K\big)(x) = C_K|x|^{-4}\big[1+\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\big(a_K+24\log\frac{|x|^2}{\ell^2}\big)\big]$ *and, by Theorem 10.7 with $D_1(O_\xi) = -48\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,O_\xi$ and Lorentz covariance,* two-point$\big(O_\xi, O_{\bar\zeta}\big)(x) = C_\zeta\,(\xi, \rho(x)\bar\zeta)|x|^{-6}\big[1+\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\big(a_\zeta+24\log\frac{|x|^2}{\ell^2}\big)\big]$, *identity (a) with $O=K$ gives C_ζ as a fixed multiple of C_K and*

$$a_\zeta - a_K = -12 + \big(\text{the coefficients of } Q_1(K) \text{ along } O_\xi\big) + \big(\text{the one-loop coefficient of } Q_1(O_{\bar\zeta}) \text{ along the derivative of } K\big)$$

where -12 *is* $-\frac{1}{2}\cdot 24$ *from* $\partial_m\big(|x|^{-4}(1+24\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log|x|^2)\big)=-4x_m|x|^{-6}\big(1+\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}(-12+24\log|x|^2)\big)$*; the remaining coefficients are those of [ch07, cor:c7-Q] (engine E5). (b)* $\text{two-point}\big(\xi\cdot O_\zeta,F_n\big)+\text{two-point}\big(O_\zeta,\xi\cdot F_n\big)=0$ *with* $\xi\cdot O_\zeta=\kappa_B B_w+\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}(c'_B B_w+c_K\bar F_w)$*: since* $\text{two-point}\big(B_w,F_n\big)$ *vanishes at order g (Proposition 10.6) and its next graphs have* $b_1=3$*, i.e. relative order* $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}{}^2$ *with respect to* $\text{two-point}_{\text{tree}}\big(\bar F_w,F_n\big)$*, and* $\text{two-point}\big(O_\zeta,\{S^{(1)}(\xi),F_n\}\big)$ *has no tree part (two undotted* λ*), the identity reads, through relative order* $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$*,*

$$c_K\,\tfrac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,\text{two-point}_{\text{tree}}\big(\bar F_w,F_n\big)=-\text{two-point}\big(O_\zeta,\{S^{(1)}(\xi),F_n\}+Q_1(F_n)\big)\big|_{\text{rel. order}\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}}:$$

c_K is fixed by one-loop two-point functions of O_ζ with the weight-$\frac{7}{2}$ operators $\{S^{(1)}(\xi),F_n\}$ (whose trilinear part $([A^{ik},A_{kj}w^j]u_\alpha,\lambda^\alpha{}_i)$-type has a nonzero order-g entry with O_ζ through one Yukawa vertex) and $Q_1(F_n)$, independently of the evaluation of $Q_1(O_\zeta)$.

Proof. (a) $\{S^{(1)}(\xi),A\}=\mp\Gamma(\xi,\lambda)$ on the slot of A [ch02, eq:c2-S1Q], and $(\iota_w A$ paired with $\Gamma(\xi,\lambda)|_{W_{\mathbb{R}}^\perp})=(\lambda,\rho(A|_{W_{\mathbb{R}}^\perp})\xi)$ by [ESW, (28)]. The classical $\{S^{(1)}(\xi),O_{\bar\zeta}\}$ contains $\frac{1}{2}(\rho(dA|_{W_{\mathbb{R}}^\perp})\xi,\bar\zeta)$-terms, i.e. the derivative of K projected by $(\xi,\rho(-)\bar\zeta)$, and terms whose two-point function with K vanishes at tree level; insert in (10.7) and compare the two tensor structures. (b) Proposition 10.18 (b), Proposition 10.6, and the propagator rule. □

10.5 Graph census at relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$

Notation of the census (all written out in [census]): $D_\varepsilon(r)=\frac{1-e^{-r^2/4\varepsilon}}{4\pi^2r^2}=\int_\varepsilon^\infty(4\pi t)^{-2}e^{-r^2/4t}dt$ is the scalar factor of $P(\varepsilon,\infty)$ [ch03, prop:c3-P], $\hat D_\varepsilon(p)=e^{-\varepsilon|p|^2}/|p|^2$; every edge carries a Schwinger time $t_e\in[\varepsilon,\infty)$, every internal vertex a position integrated over $\mathbb{R}^4$; Aut fixes O_1, O_2. The Euclidean Fourier convention is that of [ch01]. The colour factor of every graph below is $\dim\mathfrak{g}$ (tree) or $\dim\mathfrak{g}\cdot\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)/(X,Y)$ (one loop) by [ch05, lem:c5-colour]; this is the origin of the unit.

Proposition 10.20 (Scalar bilinears)**.** *For $O_a=\int\omega_a(\iota_{a_1}A,\iota_{a_2}A)$, $a=1,2$, $a_1,a_2\in W_{\mathbb{R}}^\perp$ (also written (a_1,b_1), (a_2,b_2) for the two pairs), the tree part is one cycle of two $A|_{W_{\mathbb{R}}^\perp}$ edges, $|\mathrm{Aut}|=2$,*

$$\text{two-point}_{\text{tree}}\big(O_1,O_2\big)(x)=4\hbar\dim\mathfrak{g}\,\big[(a_1,a_2)(b_1,b_2)+(a_1,b_2)(b_1,a_2)\big]\,D_0(|x|)^2,\qquad D_0(r)=\frac{1}{4\pi^2r^2},\tag{10.8}$$

two-component $\frac{1}{4}\hbar\dim\mathfrak{g}\big[a_1^{ij}a_{2ij}b_1^{kl}b_{2kl}+a_1^{ij}b_{2ij}b_1^{kl}a_{2kl}\big]D_0^2$ with $a=\frac{1}{4}a^{ij}(\cdot)_{ij}$ [dictionary, bilinears]. The graphs of relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ are exactly:

(SE) one of the two lines carries a one-particle-irreducible one-loop $A|_{W_{\mathbb{R}}^\perp}$ two-point subgraph: the seagulls with an $A|_V$ and with an $A|_{W_{\mathbb{R}}^\perp}$ loop, the bubble with one $A|_V$ and one $A|_{W_{\mathbb{R}}^\perp}$ line, the gaugino loop [ch04, second table], and the vertex $-I^{CT}[\varepsilon]$ (quadratic part); $|\mathrm{Aut}|$ as the tree times the subgraph factors $2,2,1,2$;

(EX) the exchange: cubic vertices $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$ at z_1,z_2, edges O_1z_1, O_1z_2, O_2z_1, O_2z_2 ($A|_{W_{\mathbb{R}}^\perp}$) and z_1z_2 ($A|_V$); $|\mathrm{Aut}|=2$; it contains the triangle (f) of [ch05] at O_1 and at O_2;

(QU) the quartic graph: one vertex $(A|_{W_{\mathbb{R}}^\perp})^4$ at z joined to each O_a by two edges; $|\mathrm{Aut}|=4$; it contains the bubble (e) of [ch05] at O_1 and at O_2;

(OC) the trees with one vertex $-\partial_\delta I^{CT}[\varepsilon](O_a)$, $a=1,2$, *i.e.* ε*-dependence* $-\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\big[4O_a - 2(a_a,b_a)\int\omega_a(A|_{W_{\mathbb{R}}^\perp},A|_{W_{\mathbb{R}}^\perp})\big]$ *[ch05, eq:c5-dICT];*

(F) the trees with $\hbar\varphi(O_a)$.

All other graphs vanish: a cubic vertex with two legs joined (colour), the graph with O_a *joined to one cubic vertex by both legs ((b) of [ch05], colour), Yukawa or ghost vertices on two different lines (no propagator* λ*–A, none in the slot of* $A^*|_{W_{\mathbb{R}}^\perp}$*), and the finite term* $\frac{27}{16}\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int\mathrm{dvol}\,(d^*A|_V,d^*A|_V)$ *(no* $A|_V$ *line at this order).*

Proof. At relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ a graph has the two vertices O_a and either one quartic vertex or two cubic vertices of I (one more cycle), or one vertex of order $\hbar$. The vertices with $A|_{W_{\mathbb{R}}^\perp}$ legs are $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$, $\lambda\lambda A|_{W_{\mathbb{R}}^\perp}$, $(A|_V)^2(A|_{W_{\mathbb{R}}^\perp})^2$, $(A|_{W_{\mathbb{R}}^\perp})^4$, $([A,c],A^*)$ [ch01, eq:c1-I]; the enumeration of the attachments of their legs to the four legs of O_1,O_2, with connectedness and the propagator components of [ch03, prop:c3-P], gives the list; the vanishing ones are the colour statements of [ch05, lem:c5-colour] and the absence of the components $A\otimes\lambda$, $A^*|_{W_{\mathbb{R}}^\perp}\otimes(\cdot)$. Automorphisms: (EX) the exchange $z_1\leftrightarrow z_2$ with the lines; (QU) the two pairs of parallel edges. □

Proposition 10.21 (Weight $\frac{5}{2}$: O_ζ and partners)**.** *For* O_ζ, $O_{\bar\zeta}$ *of Proposition 10.18 (both Konishi descendants by the projection* $W\otimes\wedge^2W^*\to W^*$*; the partners in the* **20′** *multiplet are* $\int\omega(\lambda_\alpha{}^i,A_{jk})$ *projected to the* **20***), the tree part is the cycle of one* λ *edge and one* $A|_{W_{\mathbb{R}}^\perp}$ *edge,* $|\mathrm{Aut}|=1$*, with value* $+24\hbar\dim\mathfrak{g}\,(\zeta,\rho(x)\bar\zeta)\,(16\pi^4)^{-1}|x|^{-6}$ *in the convention of Definition 10.1 (coefficient of* t_2t_1*), with the* $\lambda\otimes\lambda$ *component of [ch03, prop:c3-P] contracted as* $\hbar\partial_P\big((\lambda(y),g)(\lambda(x),f)\big)=\hbar\big(\rho(\partial_y)\frac{1}{4\pi^2|y-x|^2}g,f\big)$ *times the inverse pairing of* $\mathfrak{g}$ *[conj, prop:RP], and* $\sum_w\rho(w)\rho(x)\rho(w)=-6\rho(x)$ *over an orthonormal basis of* $W_{\mathbb{R}}^\perp$*; with the inversion of [ch08, prop:c8-C] (spinor factor* $-\rho(x)/|x|$ *on* λ*) the radial value is* $+24\hbar\dim\mathfrak{g}\,(\zeta',\bar\zeta)/(16\pi^4)$*, positive, and the coefficient of* t_1t_2 *would give* -24 *[ip, checks C12]. Relative order* $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$*: (*$SE_\lambda$*) the* λ *self-energies with an* $A|_V$ *and an* $A|_{W_{\mathbb{R}}^\perp}$ *line, and* $-4\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\log\varepsilon\int\mathrm{dvol}\,(\lambda,\not d\lambda)$*; (*$SE_A$*) as in Proposition 10.20; (*EX_V*)* $A|_V$ *exchanged between the* λ *line (vertex* $\lambda\lambda A|_V$*) and the* $A|_{W_{\mathbb{R}}^\perp}$ *line (vertex* $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$*); (*$EX_Y$*) the crossed Yukawa graph: Yukawa vertices at* z_1,z_2*,* O_1*'s* λ *to* z_1*,* O_1*'s A to* z_2*,* O_2*'s* $\bar\lambda$ *to* z_2*,* O_2*'s A to* z_1*, a* λ *edge* z_1z_2*; (OC), (F). No quartic vertex has fermion legs. The subdivergences at* O_a *are the one-loop graphs with* O_ζ *on the cycle and legs* $(\lambda,A|_{W_{\mathbb{R}}^\perp})$*, whose* $\log\varepsilon$ *coefficient fixes* $Z(O_\zeta)$*; Theorem 10.7 and* $D_1(O_\zeta)=-48\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,O_\zeta$ *predict the coefficient* $24\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ *of* $\log|x|^2$*, a check of the census.*

Proposition 10.22 (Weight 3: the Konishi-anomaly sector)**.** *For* $\bar F_w$*,* F_n *(two-component* $\int\omega(\bar\lambda_{\dot\alpha i}w^i,\bar\lambda^{\dot\alpha}{}_jw^j)$*,* $\int\omega(\lambda^{\alpha i}n_i,\lambda_\alpha{}^jn_j)$*) the tree part is a cycle of two* λ *edges,* $|\mathrm{Aut}|=2$*, value* $\propto\hbar\dim\mathfrak{g}\,(n_iw^i)^2|x|^{-6}$*; for* B_w*,* $\bar B_n$ *the tree part has three* $A|_{W_{\mathbb{R}}^\perp}$ *edges,* $|\mathrm{Aut}|=3!$ *modulo the colour antisymmetry, and is of relative order* $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ *with respect to the former (one bracket in each operator). The order-g entries* two-point$\big(B_w,F_n\big)$*,* two-point$\big(\bar F_w,\bar B_n\big)$ *vanish (Proposition 10.6); the graph of* $\bar F_w$*, a Yukawa vertex, an* $A|_{W_{\mathbb{R}}^\perp}$ *edge, a Yukawa vertex and* F_n *vanishes by the colour trace of* $\bar F_w$ *at a cubic vertex. At relative order* $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$*:* λ *self-energies,* $A|_V$ *and* $A|_{W_{\mathbb{R}}^\perp}$ *exchange between the two* λ *lines, (OC) with the component of* $\partial_\delta I^{CT}[\varepsilon](\bar F_w)$ *with legs* $(\bar\lambda,\bar\lambda)$*, and (F), for* $(\bar F_w,F_n)$*. The entry* two-point$\big(B_w,F_n\big)$ *at relative order* $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ *consists of the graphs with one cubic vertex (*$3+2+3=8$ *half-edges, four edges); they vanish: either the cubic vertex is the Yukawa vertex carrying both fermion legs of* F_n *and one leg of* B_w*, with colour factor* $\delta_{ac}f_{abc}=0$ *(and the two remaining legs of* B_w *joined,* $([e_a,e_b],e_c)\delta_{bc}=0$*), or it carries two scalar legs of* B_w *and one fermion leg, and no cubic vertex has this field content (Proposition*

10.6); $3+2+3+3$ and $3+2+4$ are odd, so there is no graph with two cubic vertices or one quartic vertex; the first nonvanishing classes have $b_1 = 3$ (one cubic and one quartic vertex, or three cubic vertices), relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}^2$ [ip, checks C14]. The length-changing entries of relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}$ in this sector are those of O_ζ with the trilinear weight-$\frac{7}{2}$ operators of $\{S^{(1)}(\xi), F_n\}$ through one Yukawa vertex (Proposition 10.19 (b)), each with the length-changing component of $\partial_\delta I^{CT}[\varepsilon]$ of the trilinear (legs $(\lambda, A|_{W_{\mathbb{R}}^\perp})$, superficial degree 1).

Lemma 10.23 (Numerator reduction with the heat-kernel regulator)**.** *For an edge of momentum k, $|k|^2 \hat{D}_\varepsilon(k) = e^{-\varepsilon|k|^2}$, the Fourier transform of the scalar factor $(4\pi\varepsilon)^{-2} e^{-r^2/4\varepsilon}$ of K_ε. Hence every scalar product in a numerator that is a combination of squared edge momenta reduces a graph exactly, at every ε, to graphs in which that edge is replaced by the kernel K_ε (a contact of range $\sqrt{\varepsilon}$).*

Lemma 10.24 (Localisation constants)**.** *Let a graph differ from a reference graph by replacing a subgraph γ at a vertex O_a by a subgraph γ' with the same superficial degree 0 and the same $\log\varepsilon$ coefficient. Then the difference of the two graphs tends, at $x \neq 0$, to the reference tree at O_a times the constant*

$$\lim_{R\to\infty} \Big(\int_{|w|<R} (\text{value of } \gamma' \text{ at } \varepsilon = 1) - \int_{|w|<R} (\text{value of } \gamma \text{ at } \varepsilon = 1) \Big),$$

an integral over the rescaled positions $w = z/\sqrt{\varepsilon}$ of the vertices of γ. In particular, for the bubble $D_\varepsilon(w_1) D_\varepsilon(w_2)$ with its two ends joined by K_ε instead of identified,

$$\int \frac{d^4k}{(2\pi)^4} \frac{e^{-2|k|^2}\big(e^{-|k|^2} - 1\big)}{|k|^4} = \frac{1}{8\pi^2} \int_0^\infty \frac{e^{-3k^2} - e^{-2k^2}}{k}\, dk = \frac{\log(2/3)}{16\pi^2}.$$

Proof. The difference of the integrands is concentrated where the vertices of γ are within $O(\sqrt{\varepsilon})$ of O_a, up to tails bounded by $\varepsilon\,|z|^{-6}$ (the smearing by K_ε changes a function by ε times its Laplacian), whose integral over $|z| > \rho$ is $O(\varepsilon/\rho^2)$; there the rest of the graph is smooth and equal to the reference tree up to $O(|z|)$, which gives the stated limit by dominated convergence. The integral over all w converges because the two log coefficients agree; by Parseval it is the momentum integral displayed, $\hat{K}_1(k) = e^{-|k|^2}$, $\hat{D}_1(k) = e^{-|k|^2}/|k|^2$ [script, D]. □

10.6 The scalar bilinears: exact evaluation

Theorem 10.25 (Two-point functions of the scalar bilinears at one loop)**.** *For $O_a = \int \omega_a(\iota_{a_1} A, \iota_{a_2} A)$ as in Proposition 10.20, in the minimal scheme $\varphi = 0$ with F_0 of [ch04, thm:c4-finite], put*

$$S_{12} = (a_1, a_2)(b_1, b_2) + (a_1, b_2)(b_1, a_2), \qquad T_{12} = (a_1, b_1)(a_2, b_2).$$

Then, as $\varepsilon \to 0$ at $x \neq 0$, through relative order $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}$,

$$\text{two-point}\big(O_1, O_2\big)(x) = \frac{4\hbar \dim \mathfrak{g}\, S_{12}}{16\pi^4 |x|^4} \Big[1 + \tfrac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)} \Big(\frac{27}{2} + 4\log\frac{3}{2} \Big) \Big] + \frac{32\hbar \dim \mathfrak{g}\, T_{12}}{16\pi^4 |x|^4} \tfrac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)} \Big(\log\frac{|x|^2}{\ell^2} + \gamma_E - 3\log 2 \tag{10.9}$$

For the traceless bilinears ($T_{12} = 0$): $\text{two-point}\big(O_1, O_2\big) = \text{two-point}_{\text{tree}}\big(O_1, O_2\big) \big[1 + \frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)} \big(\frac{27}{2} + 4\log\frac{3}{2} \big) \big]$*, with no* $\log|x|^2$*. For the trace $K = \int \omega(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$ ($S_{12} = 12$, $T_{12} = 36$):*

$$\text{two-point}\big(K, K\big)(x) = \frac{48\hbar \dim \mathfrak{g}}{16\pi^4 |x|^4} \Big[1 + \tfrac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)} \Big(\frac{27}{2} + 4\log\frac{3}{2} + 24 \Big(\log\frac{|x|^2}{\ell^2} + \gamma_E - 3\log 2 \Big) \Big) \Big].$$

The mixed function of a traceless bilinear with K vanishes. Two-component: $S_{12}=\frac{1}{16}\big(a_1^{ij}a_{2ij}\,b_1^{kl}b_{2kl}+a_1^{ij}b_{2ij}\,b_1^{kl}a_{2kl}\big)$, $T_{12}=\frac{1}{16}a_1^{ij}b_{1ij}\,a_2^{kl}b_{2kl}$, $K=\frac14\int\omega(A^{ij},A_{ij})$. For $\mathfrak g=\mathfrak{su}(N)$ in the normalisation of [ch06, rem:c6-konishi] the unit is $-\frac{g^2N}{32\pi^2}$ (sanity remark only).

Proof. Write every class relative to $\text{two-point}_{\text{tree}}\big(O_1,O_2\big)=4\hbar\dim\mathfrak g\,S_{12}D_0^2$, $\tau=T_{12}/S_{12}$; all steps are exact and are reproduced in [script].
Step 1 (bubble integrals). With $t_1=s\,u$, $t_2=s(1-u)$, $s\ge\varepsilon/\min(u,1-u)$, and $\log(\varepsilon|p|^2)+\gamma_E+\log 2$ abbreviated in this proof as ℓ_2 (a local abbreviation, not a symbol of the results),

$$\iint_{[\varepsilon,\infty)^2}\frac{e^{-t_1t_2|p|^2/s}}{s^2}=1-\ell_2,\qquad\iint\frac{e^{-t_1t_2|p|^2/s}}{s^3}=\frac{1}{4\varepsilon}+|p|^2\Big(\frac{\ell_2}{6}-\frac{25}{72}\Big),\qquad\iint\frac{t_1t_2\,e^{-t_1t_2|p|^2/s}}{s^4}=-\frac{\ell_2}{6}+\frac{13}{72},$$

up to $O(\varepsilon\log\varepsilon)$, from $\int_b^\infty s^{-1}e^{-as}ds=-\gamma_E-\log(ab)+O(b)$, $\int_b^\infty s^{-2}e^{-as}ds=b^{-1}+a(\gamma_E+\log(ab)-1)+O(b)$, $\int_0^1\log(u(1-u))du=-2$, $\int_0^1\log\min(u,1-u)\,du=-1-\log 2$ and their moments; checked numerically at $\varepsilon=10^{-5}$ [script, A]. The $\log\varepsilon$ coefficients agree with [ch04, lem:c4-integrals].
Step 2 (self-energy). The bubble with one $A|_V$ and one $A|_{W_{\mathbb R}^\perp}$ line has numerator $-\operatorname{tr}_{\mathfrak g}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2})(w_1,w_2)|l-p|^2$ (vertices [ch05, eq:c5-cubic], the factor $\frac12\cdot2$ of the two internal assignments of [ch04, two-point graphs]); with the centre $-\frac{t_2}{s}p$ and positive Laplacian -8 [ch04, lem:c4-gauss] its plane-wave value is $\frac{\hbar\operatorname{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}(w_1,w_2)(X_1,X_2)\big[-\frac{1}{2\varepsilon}+|p|^2(2\ell_2-\frac{13}{8})\big]$, matching both Step-1 entries of [ch04] ($-\frac14\varepsilon^{-1}$, $+1\log\varepsilon$). The gaugino loop has numerator $c\,(l,l+p)$ by $\operatorname{tr}_\Sigma(\rho(a)\rho(w)\rho(b)\rho(w'))=-16(a,b)(w,w')$ for $a,b\in V$, $w,w'\in W_{\mathbb R}^\perp$; its value $c\,[\frac{1}{2\varepsilon}+|p|^2(\frac{\ell_2}{2}-\frac78)]$ matches the Step-1 entries $-2\varepsilon^{-1}$, $-2\log\varepsilon$ for the single value $c=-8$ (two conditions, one number). The seagulls give exactly $9\varepsilon^{-1}$. Subtracting $I^{CT}[\varepsilon]$ (plane wave $\frac92\varepsilon^{-1}-2|p|^2\log\varepsilon$), the renormalised self-energy is $\frac{\hbar\operatorname{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}(w_1,w_2)(X_1,X_2)|p|^2\big(\frac{43}{8}-2(\log|p|^2+\gamma_E+\log2)\big)$, and the line is multiplied by $1+2\frac{\hbar\operatorname{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\big(\frac{43}{8}-2(\log|p|^2+\gamma_E+\log2)\big)$. By Lemma 10.4 each line becomes $D_0\big[1+\frac{\hbar\operatorname{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\big(\frac{43}{4}+4\gamma_E-12\log2+4\log|x|^2\big)\big]$; two lines:

$$\text{(SE)}=\frac{\hbar\operatorname{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\Big(\frac{43}{2}+8\gamma_E-24\log2+8\log|x|^2\Big).$$

Step 3 (masters). By Lemma 10.4 with $\int\frac{d^4k}{(2\pi)^4}\hat D_\varepsilon(k)\hat D_\varepsilon(q-k)=\frac{1}{16\pi^2}\big(1-\log(\varepsilon|q|^2)-\gamma_E-\log2\big)+O(\varepsilon)$ (Step 1),

$$\int d^4z\,D_\varepsilon(|x-z|)^2D_\varepsilon(|z|)^2=\frac{D_0^2}{8\pi^2}\Big(\log\frac{|x|^2}{\varepsilon}+\gamma_E-3\log2\Big),$$
$$\int d^4z\,D_\varepsilon(|x-z|)^2D_\varepsilon(|z|)=\frac{D_0}{16\pi^2}\Big(\log\frac{|x|^2}{\varepsilon}+1+\gamma_E-3\log2\Big),$$

both confirmed by direct position-space quadrature at $\varepsilon=10^{-4}$ to 10^{-4} relative accuracy [script, D]. By Lemma 10.24, smearing the quartic point by K_ε adds $\frac{D_0^2}{8\pi^2}\log\frac23$ (two ends), and replacing one line at O_1 by K_ε in the exchange topology gives $D_0\cdot(\text{second master})+\frac{D_0^2}{16\pi^2}\log\frac23$.
Step 4 (exchange). The cubic vertices at z_1,z_2 give, with the routing k (O_1z_1), $q-k$ (O_1z_2), k' (z_1O_2), $q-k'$ (z_2O_2), $k-k'$ (z_1z_2), the numerator $-\frac12(k+k')\cdot(2q-k-k')$ times $2^2(-\operatorname{tr}_{\mathfrak g}(\mathrm{ad}_{X_1}\mathrm{ad}_{X_2}))$ and the flavour factor of the tree: so the exchange is $\text{two-point}_{\text{tree}}\big(\Gamma^{(f)}(O_1),O_2\big)$ with $\Gamma^{(f)}(O_1)=O_1\cdot32\pi^2\frac{\hbar\operatorname{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int\frac{d^4k}{(2\pi)^4}(k+k')\cdot(2q-k-k')\hat D\hat D\hat D$, whose $\log\varepsilon$ coefficient $+2\frac{\hbar\operatorname{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ is [ch05, ex:c5-f]. The exact identity

$$(k+k')\cdot(2q-k-k')=2|q|^2-|k|^2-|q-k|^2-|k'|^2-|q-k'|^2+|k-k'|^2$$

and Lemma 10.23 split the exchange into $2|q|^2$ times the finite five-line graph (a function of $\varepsilon|q|^2$ times $|q|^{-2}$ with a limit, a contact term at $x \neq 0$ by Lemma 10.4), four graphs with one O_a-line replaced by K_ε, and the quartic topology with the point smeared. By Step 3,

$$(\mathrm{EX}) = 2\Big[\Big(2\log\frac{|x|^2}{\varepsilon} + 2\gamma_E - 4\log 2 - 2\log 3\Big) - 4\Big(\log\frac{|x|^2}{\varepsilon} + 1 + \gamma_E - 2\log 2 - \log 3\Big)\Big]\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}.$$

Step 5 (quartic). By [ch05, ex:c5-e] the bubble at O_1 is $2\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)} \cdot 16\pi^2\hat{B}_\varepsilon\big[(a_1,b_1)\int\omega_1(A|_{W^\perp_{\mathbb{R}}},A|_{W^\perp_{\mathbb{R}}}) - O_1\big]$, $\hat{B}_\varepsilon$ the bubble of Step 3; contracted with O_2,

$$(\mathrm{QU}) = 4(2\tau-1)\Big(\log\frac{|x|^2}{\varepsilon} + \gamma_E - 3\log 2\Big)\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}.$$

Step 6 (counterterms of the operators). By [ch05, eq:c5-dICT], $(\mathrm{OC}) = -8(1-\tau)\log\varepsilon\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$. Sum. The coefficient of $\log\varepsilon$ is $4+(4-8\tau)-8(1-\tau)=0$; that of $\log|x|^2$ is $8-4+(8\tau-4)=8\tau$; the constant is $\frac{43}{2}-8+8\tau(\gamma_E-3\log 2)+(8-4-4)\gamma_E-4\log 2+4\log 3$. This is (10.9). For K, $\tau=3$. □

Corollary 10.26 (Checks)**.** *(i) The coefficient* $8\tau\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$ *of* $\log|x|^2$ *equals* $-\frac{1}{4}\big(\text{two-point}_{\text{tree}}\big(D_1(O_1),O_2\big)+\text{two-point}_{\text{tree}}\big(O_1,D_1(O_2)\big)\big)/\text{two-point}_{\text{tree}}\big(O_1,O_2\big)$ *with* $D_1(O) = 4(Z(O)-4\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,O) = -8\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}(a,b)\int\omega$ (10.3)*, for every pair* (a_a,b_a)*, not only for the traceless and trace projections: Theorem 10.7 is verified on the computed graphs. (ii)* γ_E *cancels in the traceless function, as it must for an operator with* $D_1=0$*: every* γ_E *enters with a* $\log(|x|^2/\varepsilon)$*. (iii) The trace function has the Konishi exponent* $|x|^{-4-2\cdot(-24)\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}}$ *and the natural reference point* $|x|^2 = 8e^{-\gamma_E}\ell^2$*, where its logarithm vanishes. (iv) The special conformal identity of Example 10.17 holds with* $\gamma=0$*, resp.* $-24\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$*.*

Remark 10.27 (Scheme dependence of the result)**.** By Proposition 10.10: a field rescaling $\phi = a\int(A|_{W^\perp_{\mathbb{R}}},A^*|_{W^\perp_{\mathbb{R}}})$ adds $4a$ to both constants; the β-direction $\beta_2\int\mathrm{dvol}\,(dA|_{W^\perp_{\mathbb{R}}},dA|_{W^\perp_{\mathbb{R}}})$ in F_0 adds $-8\beta_2$; a finite part $\hbar\varphi(O)$ equal to a number times O (separately on the $\mathbf{20'}$ and on K) adds twice that number, divided by the unit; the unit ℓ moves the reference point. The constants $\frac{27}{2}+4\log\frac{3}{2}$ and $\frac{27}{2}+4\log\frac{3}{2}+24(\gamma_E-3\log 2)$ are therefore scheme data of the minimal UV-boundary scheme. The coefficient 24 and the vanishing for the traceless bilinears are invariant. The normalisation of the $\mathbf{20'}$ fixed by symmetry (its Ward identities with the protected multiplet, Corollary 10.11 (iii)) fixes that number on the $\mathbf{20'}$; with the target statement that the protected normalised two-point function has no order-g^2 correction (a literature statement, recorded with its regulator in [census]), it is $\hbar\varphi(O) = -\frac{1}{2}\big(\frac{27}{2}+4\log\frac{3}{2}\big)\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\,O$ in a scheme without field rescaling.